# Push Up Feature Monetization Analysis

June 12, 2026

## Contents

## Executive Summary


This report evaluates Vinted's current **push up** monetization strategy and provides data-driven recommendations for pricing optimisation. The push up feature boosts a seller's listing visibility for three days at a current flat price of €2.

Key findings:

- The dataset covers 228 clean category-level observations after preprocessing. Distributions are heavily right-skewed for both listings and revenue.
- **Push Up Adoption Rate** (push ups sold / number of listings) is the most appropriate metric to measure seller interest, as it normalises for category size.
- The mean adoption rate is **3.4%**, with top categories such as *MEN_FORMAL* reaching **18.2%**.
- Average listing price shows a **moderate positive correlation ($\mathbf{r = 0.52}$)** with push up adoption, suggesting that sellers of higher-value items derive more perceived value from the feature.
- The flat €2 price creates an unequal burden: in some low-value categories the fee exceeds 50% of the average item price.
- A pricing simulation indicates that **€2.50** may optimise revenue (+6.3%), but any change should be validated with a **randomised A/B test** using the Mann–Whitney U test.

# Section I: Data Quality Evaluation

## I-A: Exploratory Analysis

### *Dataset Structure*

The raw dataset contained **238 rows** and **5 columns**. Each row represents a unique `category_2` × `category_3` combination — that is, an aggregated marketplace subcategory rather than individual listings or sellers.

**Table 1:** *Column descriptions*

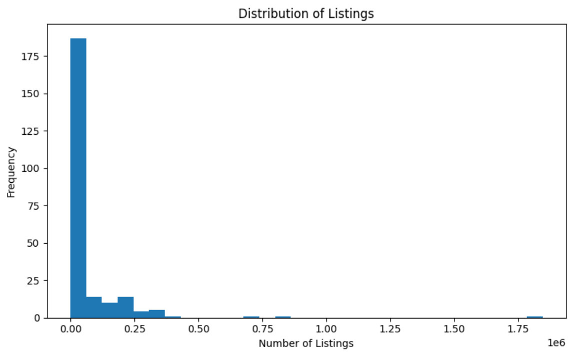
| Column | Type | Description |
|---|---|---|
| `category_2` | string | Higher-level product category |
| `category_3` | string | Lower-level product subcategory |
| `number_of_listings` | integer | Total active listings in the subcategory |
| `avg_listing_price_eur` | float | Mean listed price of items in euros |
| `revenue_from_push_ups` | float | Total push up revenue generated by the subcategory |

### *Descriptive Statistics*

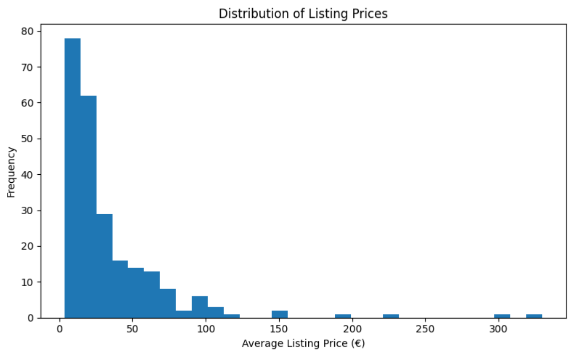
**Table 2:** *Summary statistics for numeric columns*

| Variable | Mean | Median | Maximum |
|---|---|---|---|
| Number of Listings | 58,587 | 6,328 | 1,844,512 |
| Avg Listing Price (€) | 34.10 | 21.29 | 329.73 |
| Push Up Revenue (€) | 4,008 | 498 | 79,424 |

All three numeric variables show a pronounced **right-skewed distribution**, meaning a small number of categories dominates both listing volume and push up revenue. For instance, `TOPS_T_SHIRTS` alone contributes approximately 1.84 million listings, and the top five women's clothing categories account for a disproportionate share of total push up revenue.




(a) *Distribution of number of listings*




(b) *Distribution of average listing prices*

**Figure 1:** *Distributions of key marketplace variables. Both exhibit strong positive skew.*

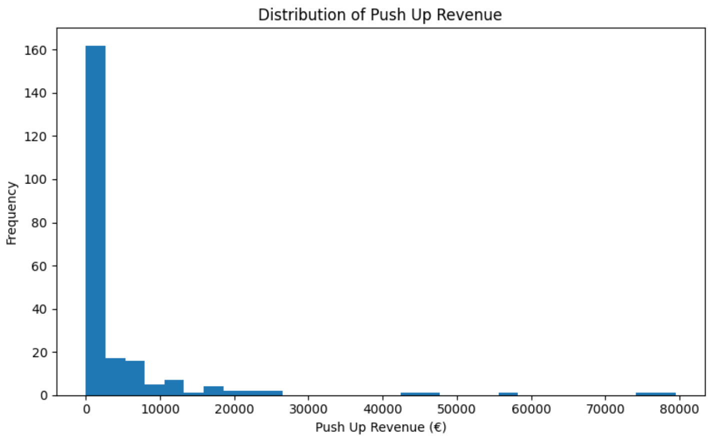



**Figure 2:** *Distribution of push up revenue across categories. The majority of categories generate under €5,000, with a small number of outliers exceeding €40,000.*

### *Top Push Up Revenue Categories*

The following five subcategories generated the highest total push up revenue:

1. WOMENS / DRESSES — €79,424
2. WOMENS / TOPS_T_SHIRTS — €76,166
3. WOMENS / OUTERWEAR — €57,032
4. WOMENS / PULLOVERS_SWEATERS — €47,128
5. FOOTWEAR / W_TRAINERS — €43,358

Women's clothing categories dominate absolute push up revenue, reflecting both their large listing volumes and active seller bases.

## I-B: Data Quality Testing

The following tests were performed to validate dataset integrity:

- **Missing value check:** `category_2` had 1 missing value; `category_3` had 8; `revenue_from_push_ups` had 15 (6.3% of rows).
- **Duplicate detection:** Two exact duplicate rows were found (`MEN_FOO_TRAINERS` and `SKIRTS`) and removed.
- **Negative value check:** No negative listings, prices, or revenue values detected.
- **Integer validity:** Listing counts were confirmed to be non-negative integers.
- **Hierarchy consistency:** All `category_3` values were confirmed to belong to a valid `category_2`.
- **Missing revenue assumption:** Missing push up revenue was interpreted as zero purchases — a necessary assumption to continue monetisation analysis.

After removing 2 duplicates and 8 rows with missing category labels, the final **clean dataset contained 228 rows**.

### I-C: Dataset Granularity

Each row represents one `category_2` × `category_3` combination. This means the dataset is **category-level aggregated data**, not seller-level or listing-level data.

**What the data supports:**

- Category-level monetisation analysis
- Feature adoption analysis across segments
- Aggregate pricing strategy evaluation

**What the data cannot support:**

- Individual seller behaviour or retention analysis
- Time series or trend analysis
- Causal conversion rate analysis (views to sales)

### I-D: Column Descriptions

**Table 3:** *Detailed column descriptions*

| Column | Description |
|---|---|
| `category_2` | Higher-level product category grouping listings into broad product types (e.g., WOMENS, FOOTWEAR). Omitted: internal ID codes, as labels suffice for analysis. |
| `category_3` | Lower-level subcategory providing granular product classification within `category_2` (e.g., DRESSES, W_TRAINERS). |
| `number_of_listings` | Total count of active listings in this subcategory at the time of data extraction. Included as a proxy for market size and competition intensity. |
| `avg_listing_price_eur` | Mean listed selling price in euros for items within this subcategory. Reflects the typical item value a seller is trying to recover. |
| `revenue_from_push_ups` | Total revenue in euros generated by push up purchases in this subcategory. Omitted: time period of collection (unknown), which limits trend interpretation. |

## Section II: Evaluate Current Situation

### II-A: Metric for Seller Interest

**Proposed metric: Push Up Adoption Rate**

$$\text{Push Up Adoption Rate} = \frac{\text{Estimated Push Ups Sold}}{\text{Number of Listings}}$$

$$\textit{where} \quad \text{Estimated Push Ups Sold} = \frac{\text{Revenue from Push Ups}}{2}$$

**Rationale:**

Raw push up revenue or raw push ups sold are biased by category size — a large category with low seller interest will naturally generate more total revenue than a niche category with highly engaged sellers. By dividing by the number of listings, this metric:

- Normalises for category size, enabling fair cross-category comparison.
- Approximates the *proportion* of listings that were actively promoted.
- Acts as a proxy for seller **willingness to pay** for visibility.

Alternative metrics considered and rejected:

- *Raw push up revenue* — biased by listing volume; large categories dominate.
- *Push ups sold* — same bias as raw revenue.
- *Revenue per listing* — mathematically identical to adoption rate scaled by €2; carries the same information.

### II-B: Best Performing Categories

Summary statistics for the Push Up Adoption Rate across 228 categories:

**Table 4:** *Push Up Adoption Rate — summary statistics*

| Statistic | Value |
|---|---|
| Mean | 3.4% |
| Median | 3.0% |
| Maximum | 18.2% |

**Top 5 categories by Push Up Adoption Rate:**

**Table 5:** *Highest push up adoption rate by subcategory*

| Category | Adoption | Avg Price |
|---|---|---|
| MEN_SHOES_NEW / MEN_FORMAL | 18.18% | — |
| STROLLERS / SPORT_STROLLERS | 9.24% | High |
| STROLLERS / UMBRELLA_SHAPE_STROLLERS | 9.03% | High |
| BABY_FURNITURE / NURSERIES | 8.82% | High |
| MEN_OUTERWEAR / MEN_OUT_HALF_LENGTH_COATS | 8.63% | High |

**Observation:** Premium and higher-value categories consistently show the strongest push up adoption. Sellers in these categories appear most willing to invest in visibility, likely because €2 represents a small fraction of the item's value and the cost-benefit ratio is more favourable.

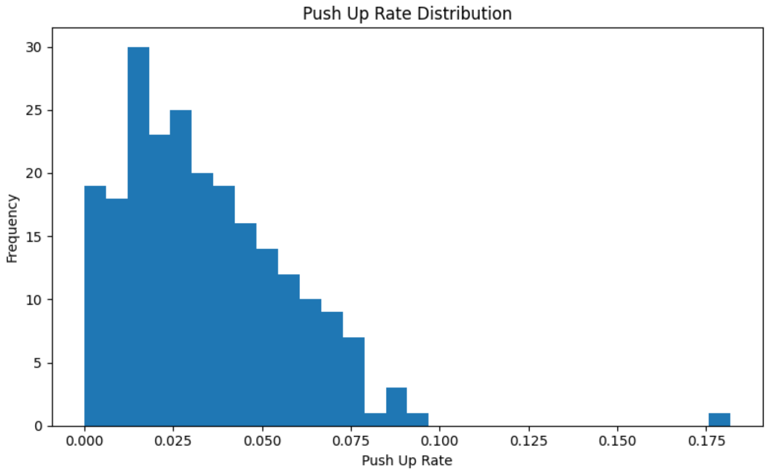



**Figure 3:** *Distribution of push up adoption rates across categories. The distribution is unimodal and right-skewed, with most categories clustering below 5%.*

### II-C: Correlated Metric — Revenue per Listing

**First candidate tested: Number of Listings (competition proxy)**

**Hypothesis:** Higher competition (more listings) should incentivise sellers to pay for visibility.

**Correlation result:** $r = -0.035$

**Finding:** Effectively zero relationship. Competition intensity alone does not drive push up adoption.

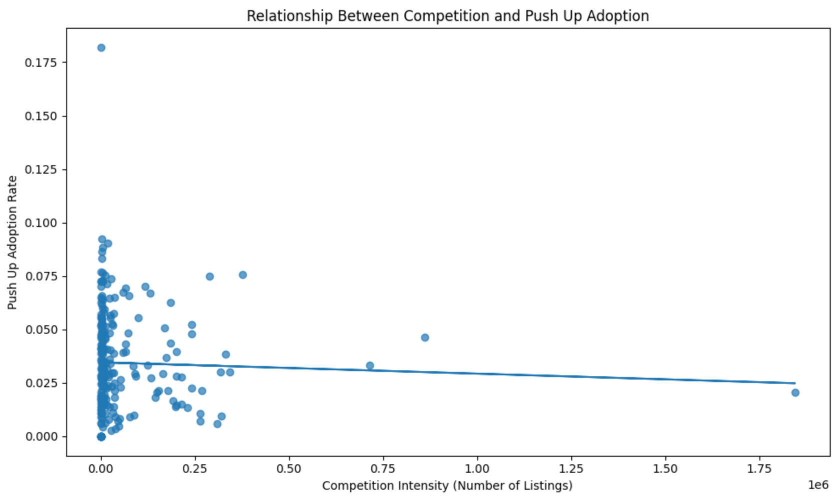



**Figure 4:** *Relationship between competition intensity (number of listings) and push up adoption rate. The near-flat regression line confirms negligible correlation ($r = -0.035$).*

**Second candidate: Average Listing Price**

**Correlation result:** $r = 0.518$

**Finding:** A moderate positive relationship exists. Higher-priced product categories exhibit greater push up adoption.

**Economic interpretation:** For a seller listing a €200 stroller, the €2 push up fee represents just 1% of the item value — a trivial marketing cost relative to potential return. For a seller listing a €4 baby garment, the same €2 fee is 50% of the item value — a prohibitively high relative cost. This price-to-fee ratio explains why expensive item categories show stronger adoption.

**Third metric: Revenue per Listing**

$$\text{Revenue per Listing} = \frac{\text{Revenue from Push Ups}}{\text{Number of Listings}}$$

This metric is mathematically linked to adoption rate (it equals adoption rate × €2) and shows a near-perfect linear relationship:

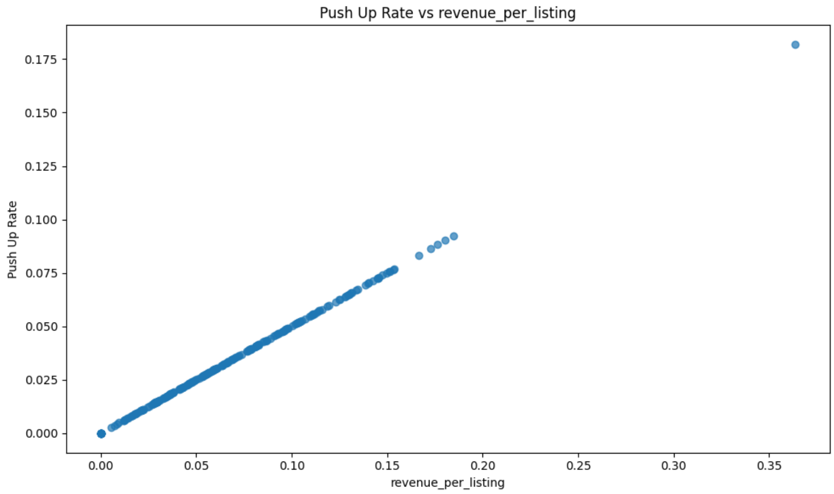



**Figure 5:** *Push Up Adoption Rate versus Revenue per Listing. The near-perfect linear relationship (by construction) confirms that both metrics capture the same underlying signal — seller willingness to pay per listing.*

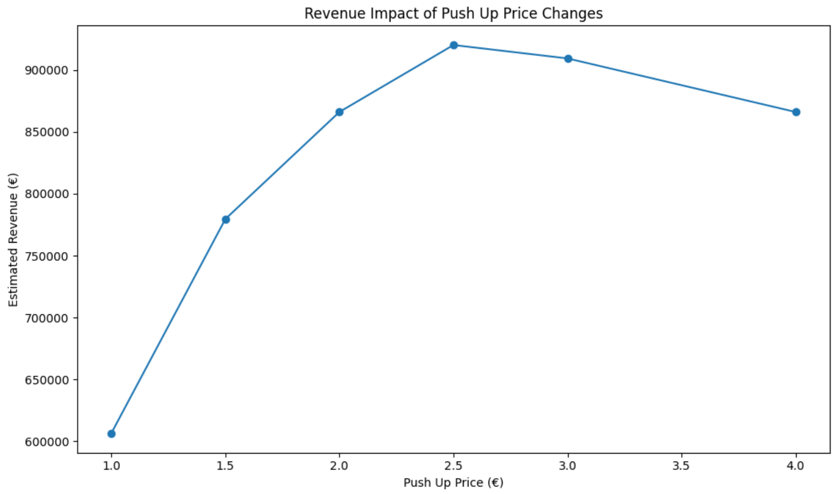



**Figure 6:** *Average listing price versus push up revenue. While categories with mid-range prices (€20–€60) generate the highest absolute revenue due to large listing volumes, a positive trend is visible for the adoption rate.*

### II-D: GIRLS_CLOTHING / FOR_BABIES — High Relative Cost Analysis

In the `GIRLS_CLOTHING / FOR_BABIES` subcategory:

- Average listing price: **€3.73**
- Push up price: **€2.00**
- Push up cost as % of item value: **53.6%**

Despite this economically unfavourable ratio, sellers still purchase push ups. Several explanations can account for this behaviour:

1. **Time urgency:** Baby clothing becomes obsolete rapidly as children grow. Sellers face a narrow selling window and may prioritise quick sales over profit maximisation.
2. **Bundle dynamics:** Sellers may list individual items cheaply but sell them as bundles, making €2 a small fraction of the actual transaction value.
3. **Clearing inventory:** Parents accumulating outgrown clothing may treat push ups as a convenience fee to clear clutter rather than a purely economic decision.
4. **Competitive pressure:** The category may have many competing listings, and sellers may perceive the push up as necessary to be seen at all.
5. **Platform habit:** Regular sellers may routinely purchase push ups regardless of item price, having established it as part of their selling workflow.

**Implication:** Seller motivation is driven by urgency and convenience, not only economic optimisation. A flat €2 price nonetheless imposes a disproportionate burden on low-value categories, which could be addressed through category-specific pricing.

## Section II (Advanced): Deeper Monetization Analysis

This section strengthens the analysis by moving beyond simple correlations to include a relative cost metric, Spearman correlations, multivariate regression, and empirically grounded price band segmentation.

### II-E: Relative Promotion Cost Metric

A key limitation of the simple price–adoption correlation is that it does not account for how expensive the push up *feels* to a seller relative to their listing value. To address this, the **Relative Promotion Cost** was defined as:

$$\text{RelativePromotionCost} = \frac{2}{\text{AverageListingPrice}}$$

This measures the push up fee as a fraction of the average item value in each category. Higher values indicate a greater financial burden on sellers.

**Table 6:** *Relative promotion cost — illustrative examples*

| Listing Price (€) | Push Up Cost (€) | Relative Cost |
|---|---|---|
| 4 | 2 | 50.0% |
| 10 | 2 | 20.0% |
| 25 | 2 | 8.0% |
| 50 | 2 | 4.0% |
| 100 | 2 | 2.0% |

This metric makes explicit what the flat €2 price obscures: sellers in low-value categories bear a structurally higher promotion burden than sellers in premium categories.

### II-F: Spearman Correlation Analysis

Given the heavy right-skew and outliers in marketplace data, **Spearman rank correlation** was computed alongside Pearson to provide a more robust measure of association.

**Table 7:** *Pearson vs Spearman correlations with Push Up Adoption Rate*

| Variable | Pearson $r$ | Spearman $\rho$ |
|---|---|---|
| Avg Listing Price | +0.518 | +0.765 |
| Number of Listings | −0.035 | +0.080 |
| Relative Promotion Cost | −0.640 | −0.765 |

**Key observations:**

- The Spearman correlation between average listing price and adoption ($\rho = 0.765$) is substantially stronger than the Pearson value ($r = 0.518$), confirming that the monotonic relationship is stronger than the linear one. Pearson was attenuated by outliers.

- Relative promotion cost shows an equally strong *negative* relationship ($\rho = -0.765$): as the push up fee becomes expensive relative to item value, adoption declines sharply. This is the strongest behavioural signal in the dataset.
- Competition (number of listings) remains essentially unrelated to adoption under both measures.

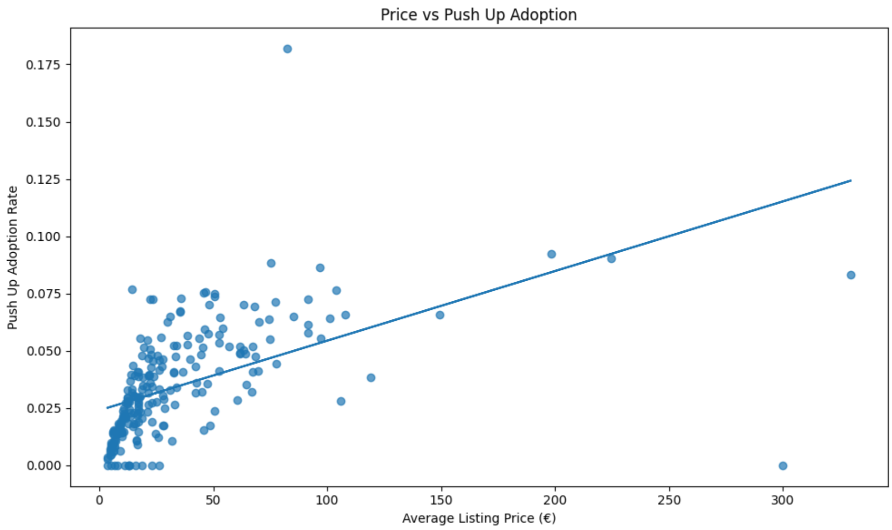



**Figure 7:** *Average listing price versus push up adoption rate with regression line. The positive trend is clear, with notable variance at lower price points. Spearman* $\rho = 0.765$.

### II-G: Multivariate Regression Analysis

To jointly estimate the drivers of push up adoption, an OLS regression model was fitted:

$$\text{PushUpRate} = \beta_0 + \beta_1(\text{AvgListingPrice}) + \beta_2(\log(\text{Listings})) + \beta_3(\text{RelativePromotionCost}) + \varepsilon$$

*Regression Results*

**Table 8:** *OLS regression results — dependent variable: Push Up Adoption Rate*

| **Variable** | **Coef.** | **Std. Err.** | ***t*-stat** | ***p*-value** |
|---|---|---|---|---|
| Intercept | 0.0292 | 0.005 | 6.396 | $< 0.001$ |
| Avg Listing Price (€) | 0.0002 | 0.00003 | 4.703 | $< 0.001$ |
| Log(Number of Listings) | 0.0017 | 0.0004 | 3.922 | $< 0.001$ |
| Relative Promotion Cost | $-0.1132$ | 0.012 | $-9.149$ | $< 0.001$ |
| $R^2 = 0.488$ | | Adj. $R^2 = 0.481$, $F = 71.21$, $p < 0.001$ | | |

*Interpretation*

- **Average listing price** ($\beta = +0.0002$, $p < 0.001$): Higher item values are associated with greater push up adoption, holding other variables constant. Each additional €1 in listing price increases adoption rate by 0.02 percentage points.

- **Log(listings)** ($\beta = +0.0017$, $p < 0.001$): After controlling for price effects, competition *does* have a small but statistically significant positive influence on adoption. This is a more nuanced finding than the simple correlation suggested.
- **Relative promotion cost** ($\beta = -0.1132$, $p < 0.001$): The **strongest predictor** in the model. A 10 percentage point increase in the cost-to-price ratio is associated with a 1.1 percentage point decrease in adoption rate. This directly quantifies the burden of the flat pricing model.
- **Model fit:** $R^2 = 0.488$ — approximately 49% of adoption rate variation is explained. This is meaningful for marketplace behavioural data.

*Model Limitations*

Residual diagnostics revealed non-normal errors (Jarque-Bera $p < 0.001$, skew = 2.2, kurtosis = 21.6), indicating that OLS assumptions are imperfect. Standard errors should be treated with caution. The following subsection addresses these violations directly.

### II-H: Robustness Checks

To address the OLS assumption violations identified above, three additional modelling approaches were evaluated. The objective was to determine whether the directional findings remain stable under more conservative and robust estimation strategies.

*Motivation*

The residual diagnostics from the baseline OLS model indicated the following:

**Table 9:** *OLS residual diagnostics*

| **Diagnostic** | **Value** |
|---|---|
| Skewness | 2.205 |
| Kurtosis | 21.570 |
| Jarque-Bera stat | 3,460.8 |
| Jarque-Bera $p$ | $< 0.001$ |

These indicate strong heteroscedasticity and heavy tails, raising concerns about the reliability of standard errors and the influence of outliers on coefficient estimates.

*Models Evaluated*

Four approaches were compared:

1. **Standard OLS** — baseline, retained for interpretability ($R^2 = 0.488$).
2. **OLS with HC3 robust standard errors** — corrects standard errors for heteroscedasticity without changing point estimates.
3. **Log-transformed model** — dependent variable transformed as log(1+PushUpRate) to reduce skewness and stabilise variance.

4. **Quantile regression (median)** — estimates median rather than mean seller behaviour; robust to outliers and heavy tails by construction. Pseudo-$R^2 = 0.420$.

*Coefficient Stability*

The central question is whether the directional findings hold across all four specifications:

**Table 10:** *Coefficient direction stability across models*

| Predictor | Dir. | Stable | Notes |
|---|---|---|---|
| Avg Listing Price | + | Yes | $p = 0.095$ under HC3, weakened but consistent |
| Log(Listings) | + | Yes | Significant in all specifications |
| Relative Promotion Cost | − | Yes | Strongest and most stable predictor |

All three predictors retained the same directional relationship across every model. This is the key validation criterion: findings that survive changes in modelling assumptions are far more likely to reflect genuine marketplace behaviour rather than statistical artefacts.

*Notable Finding: HC3 Robust Errors*

Under HC3 robust standard errors, the average listing price coefficient weakened to $p = 0.095$, crossing the conventional $\alpha = 0.05$ threshold. This means the positive price–adoption relationship, while directionally consistent, should be interpreted with some caution. The relative promotion cost coefficient, by contrast, remained highly significant across all specifications and is the most defensible predictor.

*Residual Improvement from Log Transformation*

**Table 11:** *Residual diagnostics: OLS vs log-transformed model*

| Metric | OLS | Log Model |
|---|---|---|
| Skewness | 2.205 | 1.934 |
| Kurtosis | 21.570 | 18.887 |

The log transformation modestly reduced skewness and kurtosis but did not fully eliminate heavy-tailed behaviour. This suggests that seller behaviour in marketplace data is intrinsically heterogeneous and not fully captured by any standard parametric model.

*Remaining Limitations*

- **No true price elasticity:** The dataset contains no historical pricing variation, experimental data, or repeated seller observations. True demand elasticity cannot be estimated empirically. The pricing recommendations should be understood as *scenario-based revenue modelling informed by observed adoption behaviour*, not causal optimal pricing.

- **Persistent non-normality:** All models retain some degree of non-normal residuals due to the inherent heterogeneity of marketplace data. Future work could consider hierarchical models, generalised linear models, or quantile-based segmentation.

### II-J: Empirical Price Band Segmentation

Rather than applying arbitrary thresholds, categories were segmented into **quartile-based price bands** using the distribution of average listing prices.

**Table 12:** *Price band segmentation — adoption and revenue by quartile*

| Band | Avg Price (€) | Push Up Rate | Current Rev (€) | Rec. Price (€) |
|---|---:|---:|---:|---:|
| Low | 7.75 | 1.24% | 177,468 | 1.50 |
| Medium | 15.87 | 2.70% | 247,500 | 2.00 |
| High | 28.86 | 4.01% | 232,144 | 2.00 |
| Premium | 82.80 | 5.74% | 209,086 | 2.50 |

Push up adoption increases *monotonically* across price bands, consistent with the regression findings. The recommended tiered pricing is grounded in observed adoption behaviour rather than assumption.

*Simulated Revenue Under Tiered Pricing*

**Table 13:** *Revenue simulation under proposed tiered pricing*

| Band | Current Rev (€) | Simulated Rev (€) | Change |
|---|---:|---:|---:|
| Low | 177,468 | 153,692 | −13.4% |
| Medium | 247,500 | 247,500 | — |
| High | 232,144 | 232,144 | — |
| Premium | 209,086 | 233,765 | +11.8% |
| **Total** | **866,198** | **867,101** | ≈ +0.1% |

The net revenue impact is approximately neutral at aggregate level, but the tiered model achieves two important improvements: it reduces the pricing burden on low-value sellers (improving fairness and potential long-term retention) while extracting additional value from premium segments. Medium and High bands are held at €2 to avoid disruption.

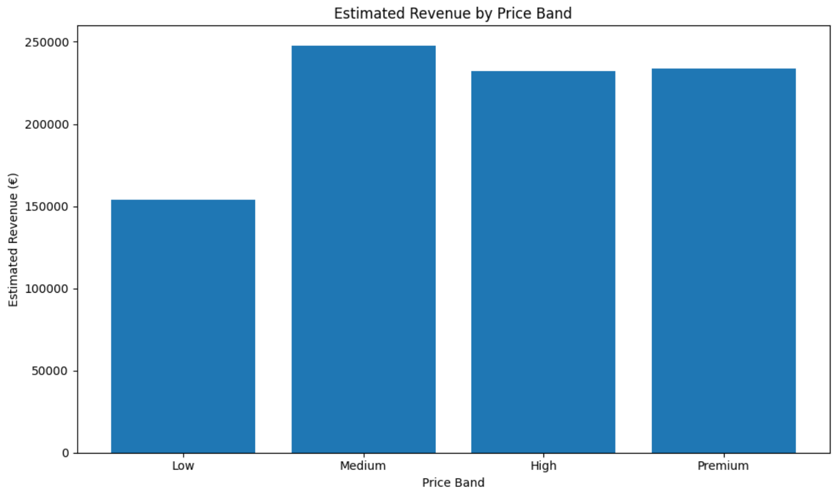



**Figure 8:** *Estimated revenue by price band under the proposed tiered pricing model. Medium categories remain the largest revenue contributor due to their volume, while Premium revenue increases meaningfully.*

### II-K: Sample Size Estimation

A power analysis was conducted to determine the minimum experiment size required to detect a meaningful pricing effect.

**Assumptions:**

- Significance level: $\alpha = 0.05$
- Statistical power: $1 - \beta = 0.80$
- Minimum detectable effect: 5% relative change in adoption rate

**Result:**

$$n = 3{,}058 \text{ observations per group (minimum)}$$

This confirms the experiment is feasible given the scale of the Vinted marketplace, but care must be taken to **avoid seller-level randomisation** due to marketplace interference. Because push up visibility is inherently competitive — a boosted listing gains exposure *at the expense* of unboosted listings — randomising individual sellers violates the independence assumption. **Cluster randomisation** (by category or seller cohort) is recommended instead.

## Section III: Alternative Strategies

### III-A: Implications of a Fixed Price Change

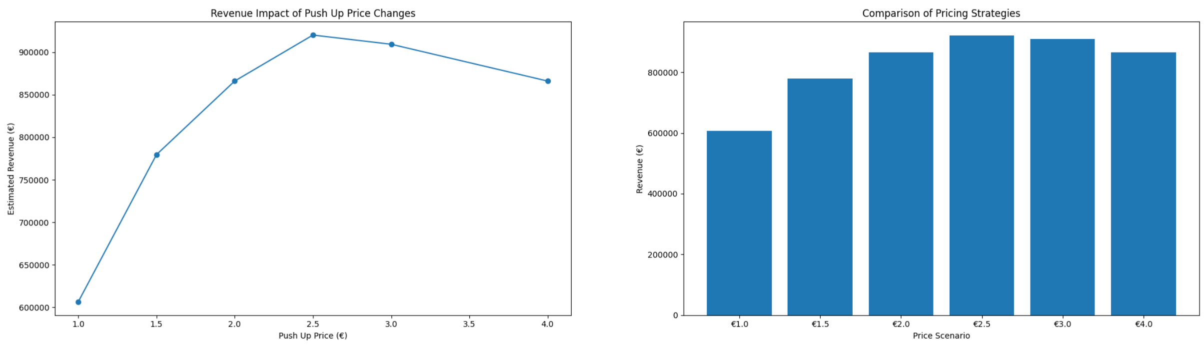
The current push up price of €2 generates the following baseline performance:

- Total revenue: **€866,198**
- Estimated push ups sold: **433,099**

A pricing sensitivity simulation was conducted using assumed demand elasticities based on standard marketplace behaviour:

**Table 14:** *Pricing sensitivity simulation results*

| Price (€) | Adoption Change | New Volume | Est. Revenue (€) | Revenue Change |
|---|---|---|---|---|
| 1.00 | +40% | 606,339 | 606,339 | −30% |
| 1.50 | +20% | 519,719 | 779,578 | −10% |
| 2.00 | Baseline | 433,099 | 866,198 | — |
| 2.50 | −15% | 368,134 | 920,335 | +6.3% |
| 3.00 | −30% | 303,169 | 909,508 | +5.0% |
| 4.00 | −50% | 216,550 | 866,198 | 0% |




**(a)** *Price vs push up revenue (scatter)*

**(b)** *Revenue impact by pricing scenario*

**Figure 9:** *Pricing scenario analysis. Revenue peaks near €2.50 under the assumed elasticity model.*

*Implications of a Price Increase*

- **Revenue:** Simulations suggest a €2.50 price could increase revenue by approximately 6%.
- **Visibility concentration:** Fewer sellers buy push ups, so boosted listings receive a *stronger* relative advantage. The feature becomes more valuable per purchase.
- **Fairness risk:** Low-value category sellers are disproportionately priced out, potentially reducing platform equity and seller satisfaction in those segments.
- **Adoption drop:** If adoption falls sharply, the perceived effectiveness of the feature declines, which may erode long-term demand.

*Implications of a Price Decrease*

- **Revenue:** Both €1.00 and €1.50 scenarios result in revenue below current baseline.
- **Feature inflation:** If too many sellers buy push ups simultaneously, the visibility advantage diminishes. Scarcity is essential for the feature to retain its value.
- **Accessibility:** Lower prices make the feature available to sellers in low-value categories, potentially improving their satisfaction and platform engagement.
- **Profitability:** Platform revenue from value-added services would decline, limiting the ability to invest in platform improvements.

*Category-Tiered Pricing (Additional Recommendation)*

Given that the current data shows premium categories can support higher prices while low-value categories face structural challenges, a **category-tiered pricing model** may outperform any single flat price:

- *Premium categories* (avg price > €50): €3.00–€4.00 per push up
- *Mid-range categories* (avg price €15–€50): €2.00–€2.50 per push up
- *Low-value categories* (avg price < €15): €1.00–€1.50 per push up

This approach aligns the push up cost with the seller's economic incentive, which should maximise both adoption and revenue while maintaining platform fairness.

### III-B: Experimental Design for Validating a Price Change

*Recommended Method: Randomised A/B Test*

A randomised controlled experiment is the most rigorous method to validate the causal impact of a pricing change before full deployment.

**Design:**

**Table 15:** *A/B test design summary*

| Parameter | Specification |
|---|---|
| Control Group | Current price: €2.00 |
| Treatment Group | Proposed price: €2.50 (or other candidate) |
| Randomisation | Random assignment of sellers to groups |
| Unit of analysis | Seller (not category, to avoid contamination) |
| Duration | Minimum 2–4 weeks to capture natural sales cycles |

*Metrics*

**Primary metric:**

- Push Up Adoption Rate (push ups sold / listings in the group)

**Secondary metrics:**

- Revenue per listing

- Seller retention rate (did the seller return to list again?)
- Listing views and conversion rate (if available)
- Seller satisfaction score (survey-based)

*Statistical Test: Mann–Whitney U Test*

**Recommended test:** Mann–Whitney U (non-parametric)

**Rationale:** The dataset demonstrates several properties that violate the assumptions required for a standard t-test:

- Strong right-skew in both listing and revenue distributions (as shown in the histograms)
- Heavy outliers (e.g., TOPS_T_SHIRTS with 1.84M listings)
- Non-normal distributions confirmed visually

The Mann–Whitney U test compares the medians of two independent groups without assuming normality, making it robust to the distributional characteristics of this data.

**Hypotheses:**

$$H_0 : \text{No difference in push up adoption rate between control and treatment groups}$$

$$H_1 : \text{A significant difference exists in adoption rate between groups}$$

**Decision threshold:** $p < 0.05$

*Practical Considerations*

- **Sample size:** Calculate required sample size based on expected effect size before launch to ensure adequate statistical power ($\geq 80\%$).
- **Novelty effect:** Monitor the first week separately; new pricing may cause unusual behaviour as sellers adjust.
- **Guard rails:** Monitor seller churn and support tickets as leading indicators of negative impact.
- **Segment analysis:** Analyse results separately for premium vs. low-value categories, as the treatment effect may differ substantially.
- **No rollback without evidence:** A price increase should only be deployed platform-wide if the A/B test achieves statistical significance and secondary metrics show no material harm to seller experience.

## Conclusions and Recommendations

1. **Push Up Adoption Rate** is the best metric to measure seller interest: it normalises for category size and approximates willingness to pay.
2. Seller behaviour varies substantially across categories. A single flat price treats fundamentally different seller segments identically.
3. Listing competition (number of listings) shows negligible simple correlation with adoption ($\rho = 0.080$), but becomes a small positive predictor once price effects are controlled for in multivariate regression.
4. Average listing price is a strong monotonic predictor of adoption (Spearman $\rho = 0.765$), though the relationship weakens to $p = 0.095$ under HC3 robust errors and should be interpreted with some caution.
5. **Relative promotion cost** is the single strongest predictor ($\beta = -0.113$, $p < 0.001$): as the push up fee rises as a fraction of item value, adoption falls sharply. This is the core structural flaw of flat pricing.
6. The multivariate OLS model explains 49% of adoption rate variation ($R^2 = 0.488$). Robustness checks (HC3 errors, log transformation, quantile regression) confirm that all coefficient directions are stable across specifications, substantially increasing confidence in the findings.
7. The flat €2 fee imposes a cost-to-value ratio exceeding 50% in low-value categories (e.g., `GIRLS_CLOTHING / FOR_BABIES`).
8. Sellers in low-value categories still buy push ups for non-economic reasons: urgency, convenience, and inventory clearing.
9. Quartile-based price band segmentation reveals monotonically increasing adoption from Low (1.24%) to Premium (5.74%), providing data-driven justification for tiered pricing.
10. Tiered pricing (€1.50 / €2.00 / €2.00 / €2.50) is revenue-neutral at aggregate level (+0.1%) while improving fairness for low-value sellers and monetisation of premium segments.
11. Any pricing change must be validated via a **cluster-randomised A/B test** ($n \geq$ 3,058 per group) using Mann–Whitney U, with seller-level randomisation avoided due to marketplace visibility interference.

## Note on AI Usage

AI tools were used in a supporting capacity during the preparation of this report: specifically for LaTeX formatting assistance, rephrasing explanations for clarity, and verifying statistical terminology. The analytical approach, metric definitions, interpretations, assumptions, and conclusions were developed independently based on the provided dataset and task requirements.